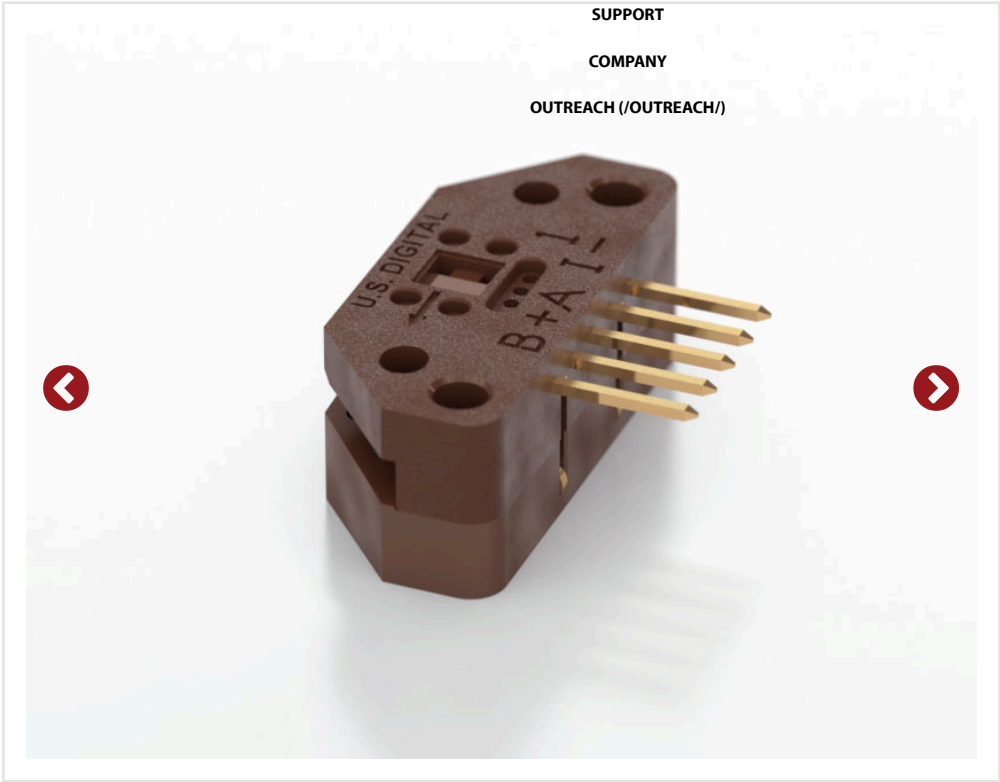


EM1

TRANSMISSIVE OPTICAL ENCODER MODULE



PHOTOS (4)

MECHANICAL DRAWINGS (3)

EM1 Features

- Encoder module for measuring linear or rotary movement
- Two-channel quadrature with optional index channel
- 21 Rotary Resolutions from 32 to 2,500 CPR (128 to 10,000 PPR)
- 9 Linear Resolutions from 120 to 500 LPI (x1 quadrature decoding)
- Sink/source 8 mA outputs
- Internal decoupling capacitor for enhanced noise immunity
- Single 5 VDC supply
- Improved replacement for HEDS-9000 Series

SEE MORE INFO BELOW

CONFIGURE THE EM1

EM1 Product Description

The EM1 is a transmissive optical encoder module designed to measure rotary or linear motion when paired together with an encoder disk or linear strip. The EM1 consists of a lensed LED source and a monolithic detector IC enclosed in a small polymer package. The EM1 uses phased array detector technology, allowing for a wider gap tolerance and reliable performance even with some disk contamination.

The EM1 encoder module provides digital A & B quadrature outputs with an optional index channel. Each EM1 module is resolution-specific and is matched to the resolution of an encoder disk or linear strip. The EM1 module supports 21 different resolutions. The EM1 operates with a single 5V supply and provides single-ended outputs capable of sinking and sourcing 8mA. An internal 0.1 μ F decoupling capacitor provides enhanced noise immunity.

For open collector and higher voltage applications, add the PC3 (/products/accessories/interfaces/cable-drivers/pc3/) cable driver, or for differential cable driver outputs, add the PC4 (/products/accessories/interfaces/cable-drivers/pc4/) cable driver.

Accessories such as encoder disks, linear strips, mating connectors and cables are also available.

Product Specifications

COMPATIBLE 1" & 2" DISKS	+
COMPATIBLE 1" & 2" HUBDISKS	+
COMPATIBLE LINEAR STRIPS	+
ENVIRONMENTAL	+

Product Configurator

TO VIEW PRICING CHOOSE EITHER OPTION 1 OR 2 BELOW.

Questions? Contact our support team.
(/company/contact)

Option 1: Enter the Product Part Number

Already have your part number (e.g., E5-5000-394-IE-S-D-D-3)? Enter it in the following field.

EM1-?

SET CONFIGURATION

Option 2: Build your Product

Customize your product step-by-step. Need help? Our support team is here to help. (/support)

Leave a message

OPERATING CONDITIONS						+
PRODUCTS						
ELECTRICAL SPECIFICATIONS						—
APPLICATIONS						
SUPPORT						
COMPANY						
OUTREACH (/OUTREACH/)						
PARAMETER	MIN.	TYP.	MAX.	UNITS	CONDITIONS	
Supply Voltage	4.5	5.0	5.5	V	Ripple < 100 mVpp	
Supply Current, EM1-0- (linear strip)		27 54	33 65	mA mA	LPI < 250, no load LPI ≥ 360, no load	
Supply Current, EM1-1- (1" disk)		27 54	33 65	mA mA	CPR < 500, no load CPR ≥ 500, no load	
Supply Current, EM1-2- (2" disk)		27 54	33 65	mA mA	CPR < 1000, no load CPR ≥ 1000, no load	
Low-level Output		0.05	0.5	V V	I _{OL} = 8mA max. No load	
High-level Output	2.0	4.8		V V	I _{OH} = -8mA max. No load	
Output Current Per Channel	-8		8	mA		
Load Capacitance			100	pF		
Output Rise Time		110		nS		
Output Fall Time		100		nS		
TIMING CHARACTERISTICS						—
ENCODING CHARACTERISTICS:						
- Specifications apply over the entire operating temperature range. - Values are for the worst error over full rotation. - Refer to the timing diagram below.						
PARAMETER	SYMBOL	MIN.	TYP.	MAX.	UNITS	
Symmetry	X, Y	150	180	210	°e (/support/resources/glossary/electrical-degree/)	
Quadrature	Z	60	90	120	°e (/support/resources/glossary/electrical-degree/)	
Index Pulse Width	Po	40	90	120	°e (/support/resources/glossary/electrical-degree/)	
Ch. I Rise After Ch. B or Ch. A Fall	t1	50	100	200	ns	
Ch. I Fall After Ch. B or Ch. A Rise	t2	-10	15	25	ns	
TIMING DIAGRAM:						

TYPE

Linear Strip (0)

1" Disk (1)

2" Disk (2)

RESOLUTION

INDEX

SUMMARY

Item Number

EM1-?

Quantity

1

Quantity Level

1

Unit Price

\$0.00

Total Price

\$0.00

Qty Level

Not Configured

Saved Configurations (0):

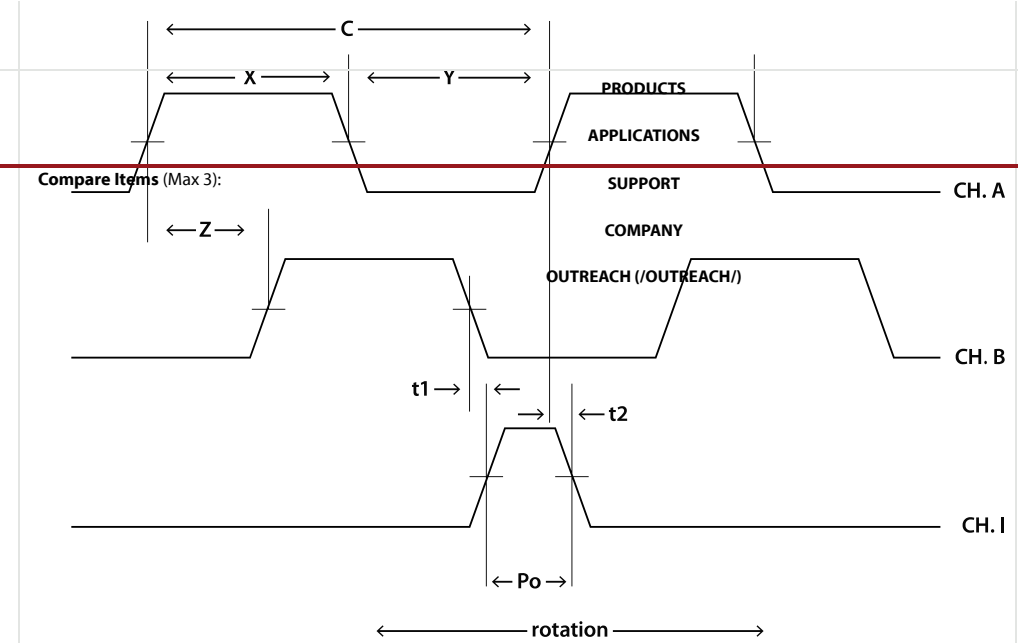
+ ADD THIS CONFIGURATION TO YOUR LIST

Configurations currently in your list (0)

Feedback

US Digital's mission is a commitment to quality and constant improvement. If you find an error to a product on this page, please let us know (mailto:website@usdigital.com?subject=Website%20Issue%20Feedback)!

Leave a message



- CPR:** The number of Cycles (C) of the A or B outputs Per Revolution.
- Cycle Error:** An indication of cycle uniformity. The difference between an observed shaft angle which gives rise to one electrical cycle, and the nominal angular increment of 1/CPR of a revolution.
- Index (I):** The index output goes high once per revolution, coincident with the low states of channels A and B, nominally 1/4 of one cycle (90 °e).
- LPI:** Lines Per Inch. The number of Cycles (C) of the A or B output per inch of linear strip movement.
- One Shaft Rotation:** 360 mechanical degrees.
- One Electrical Degree (°e):** 1/360th of one cycle.
- One Cycle (C):** 360 electrical degrees (°e). Each cycle can be decoded into 1 or 4 states, referred to as X1 or X4 resolution multiplication.
- PPR:** The number of resolvable Positions Per Revolution of the encoder disk with x4 quadrature decoding.
- Quadrature (Z):** The phase lag or lead between channels A and B in electrical degrees, nominally 90 °e.
- Symmetry:** A measure of the relationship between (X) and (Y) in electrical degrees, nominally 180°e.

INSTALLATION TORQUE	+
EM1 / HEDS COMPARISON	+
PIN-OUTS	+
PRODUCT CHANGE NOTIFICATIONS	+

Additional Information

Product Notes

- US Digital® warrants its products against defects in materials and workmanship for two years. See complete warranty (<https://www.usdigital.com/company/legal/terms-and-conditions-of-sale/#warranty>) for details.

Related

- PCN 1016 - EM1 LED Die (/support/resources/product-change-notifications/pcn-1016-em1-led-die/)
- PCN 4199 - EM1 & EM2 Update (/support/resources/product-change-notifications/pcn-4199-em1-em2-update/)

Press Releases

- 800 CPR Resolution Option Added to EM1 Module (/news/press-releases/800-cpr-resolution-option-added-to-em1-module/)

PRODUCTS

APPLICATIONS

3D Model Downloads

Compare Items (Max 3):
Please configure your product first to download a 3D model.

SUPPORT

(Note: The formats below will become links if there are 3D models available.)

COMPANY

- SolidWorks Format (/assets/models/EM1_WEB/SolidWorks/EM1_WEB_SolidWorks.zip)
- IGES Format (/assets/models/EM1_WEB/IGES/EM1-X-X-X_IGES.zip)
- Parasolid Format (/assets/models/EM1_WEB/Parasolid/EM1-X-X-X_Parasolid.zip)
- STEP Format (/assets/models/EM1_WEB/STEP/EM1-X-X-X_STEP.zip)

Datasheet

- Datasheet-EM1.pdf (/assets/datasheets/datasheet-EM1.pdf)

Products
(/products/)

Support
(/support/)

Company
(/company/)

Legal
(/company/legal/)

Contact (/company/contact/)

Encoders (/products/encoders/)	Resource Library (/support/resources/)	About Us (/company/about-us/)	Privacy Policy (/company/legal/privacy-policy/)	Toll Free: 800.736.0194 (tel:+18007360194)	1400 NE 136th Ave. Vancouver, WA 98684 USA (https://www.google.com/maps/place/1400+NE+136th+Ave,+Vancouver,+WA+98684/@49.2553628,-122.5362811,15z)
Inclinometers (/products/inclinometers/)	FAQs (/support/faqs/)	Careers (/company/careers/)	Terms and Conditions of Sale (/company/legal/terms-and-conditions-of-sale/)	Worldwide: 360.260.2468 (tel:+13602602468)	
Accessories (/products/accessories/)	Reference Documents (/support/resources/reference/)	News (/company/news/)	Terms and Conditions of Purchase (/company/legal/terms-and-conditions-of-purchase/)	Tech Support: 360.397.9999 (tel:+13603979999)	
Discontinued Products (/products/discontinued-products/)	Outreach Information and Support Videos (/support/outreach-info-and-support-videos/)	Company Videos (/company/videos/)	Payment (/company/legal/terms-and-conditions-of-sale/#payment)	Outreach: 360.816.1400 (tel:+13608161400)	Monday - Thursday 6:30 AM - 5:00 PM PDT
Applications (/applications/)	Assembly Instructions (/support/resources/assembly-instructions/)	Distributors (/company/distributors/)	Shipping and Delivery (/company/legal/terms-and-conditions-of-sale/#shipping-and-delivery)	Warranty (/company/legal/terms-and-conditions-of-sale/#warranty)	Friday (no shipping) 6:30 AM - 4:00 PM PDT
	Assembly Tutorials (/company/outreach/tutorials/outreach/)	Outreach Content (/support/site-content/)	Website Defacement (/company/legal/website-defacement/)	sales@usdigital.com (mailto:sales@usdigital.com)	BACK TO TOP
	Product Change Notifications (/support/site-content/product-change-notifications/)			support@usdigital.com (mailto:support@usdigital.com)	
	Software Downloads (/support/resources/downloads/software/)				
	Model Downloads (/support/resources/downloads/model-downloads/)				



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